

Probability & Statistics

GATE Computer Science & IT Previous Year Questions (2016 – 2025) | Total Questions: 23

Question 1 of 23 | GATE 2025 Set 1 (CS1) | Q.32 (Computer Science & IT)

Q.32

A box contains 5 coins: 4 regular coins and 1 fake coin. When a regular coin is tossed, the probability $P(\text{head}) = 0.5$ and for a fake coin, $P(\text{head}) = 1$. You pick a coin at random and toss it twice, and get two heads. The probability that the coin you have chosen is the fake coin is _____. (rounded off to two decimal places)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 2 of 23 | GATE 2025 Set 1 (CS1) | Q.56 (Computer Science & IT)

Q.56

Suppose a 5-bit message is transmitted from a source to a destination through a noisy channel. The probability that a bit of the message gets flipped during transmission is 0.01. Flipping of each bit is independent of one another. The probability that the message is delivered error-free to the destination is _____. (rounded off to three decimal places)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 3 of 23 | GATE 2025 Set 1 (CS1) | Q.58 (Computer Science & IT)

Q.58

Consider a probability distribution given by the density function $P(x)$.

$$P(x) = \begin{cases} Cx^2, & \end{cases}$$

for $1 \leq x \leq 4$

0,

for $x < 1$ or $x > 4$

The probability that x lies between 2 and 3, i.e., $P(2 \leq x \leq 3)$ is _____.

(rounded off to three decimal places)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 4 of 23 | GATE 2025 Set 2 (CS2) | Q.64 (Computer Science & IT)

Q.64

A quadratic polynomial $(x - \alpha)(x - \beta)$ over complex numbers is said to be square invariant if $(x - \alpha)(x - \beta) = (x - \alpha^2)(x - \beta^2)$. Suppose from the set of all square invariant quadratic polynomials we choose one at random.

The probability that the roots of the chosen polynomial are equal is _____.

(rounded off to one decimal place)

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 5 of 23 | GATE 2024 Set 1 (CS1 S5) | Q.27 (Computer Science & IT)

Q.27

Let A and B be two events in a probability space with $P(A) = 0.3$, $P(B) = 0.5$, and $P(A \cap B) = 0.1$. Which of the following statements is/are TRUE?

- (A) The two events A and B are independent
- (B) $P(A \cup B) = 0.7$
- (C) $P(A \cap B^c) = 0.2$, where B^c is the complement of the event B
- (D) $P(A^c \cap B^c) = 0.4$, where A^c and B^c are the complements of the events A and B, respectively

Question 6 of 23 | GATE 2024 Set 1 (CS1 S5) | Q.63 (Computer Science & IT)

Q.63

A bag contains 10 red balls and 15 blue balls. Two balls are drawn randomly without replacement. Given that the first ball drawn is red, the probability (rounded off to 3 decimal places) that both balls drawn are red is _____

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[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 7 of 23 | GATE 2024 Set 2 (CS2 S6) | Q.18 (Computer Science & IT)

Q.18

When six unbiased dice are rolled simultaneously, the probability of getting all distinct numbers (i.e., 1, 2, 3, 4, 5, and 6) is

- (A) 1/324
 - (B) 5/324
 - (C) 7/324
 - (D) 11/324
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Question 8 of 23 | GATE 2024 Set 2 (CS2 S6) | Q.42 (Computer Science & IT)

Q.42

Consider an array X that contains n positive integers. A subarray of X is defined to be a sequence of array locations with consecutive indices.

The C code snippet given below has been written to compute the length of the longest subarray of X that contains at most two distinct integers. The code has two missing expressions labelled (P) and (Q).

```
int first=0, second=0, len1=0, len2=0, maxlen=0;
for (int i=0; i < n; i++) {
```

```

if (X[i] == first) {
    len2++; len1++;
} else if (X[i] == second) {
    len2++;
    len1 = □□□□□(P)□□□□□□□□;
    second = first;
} else {
    len2 = □□□□□(Q)□□□□□□□□;
    len1 = 1; second = first;
}
if (len2 > maxlen) {
    maxlen = len2;
}
first = X[i];
}

```

Which one of the following options gives the CORRECT missing expressions?

(Hint: At the end of the i -th iteration, the value of $len1$ is the length of the longest subarray ending with $X[i]$ that contains all equal values, and $len2$ is the length of the longest subarray ending with $X[i]$ that contains at most two distinct values.)

- (A) (P) $len1+1$ (Q) $len2+1$
- (B) (P) 1 (Q) $len1+1$
- (C) (P) 1 (Q) $len2+1$
- (D) (P) $len2+1$ (Q) $len1+1$ Organizing Institute: IISc, Bengaluru

Question 9 of 23 | GATE 2024 Set 2 (CS2 S6) | Q.44 (Computer Science & IT)

Q.44

Let x and y be random variables, not necessarily independent, that take real values in the interval $[0,1]$. Let $z = xy$ and let the mean values of x, y, z be $\bar{x}, \bar{y}, \bar{z}$, respectively. Which one of the following statements is TRUE?

- (A) $\bar{z} = \bar{x}\bar{y}$
- (B) $\bar{z} \leq \bar{x}\bar{y}$
- (C) $\bar{z} \geq \bar{x}\bar{y}$
- (D) $\bar{z} \leq \bar{x}$

Question 10 of 23 | GATE 2023 Single Session | Q.52 (Computer Science & IT)

Q.52

Consider a random experiment where two fair coins are tossed. Let A be the event that denotes HEAD on both the throws, B be the event that denotes HEAD on

the first throw, and C be the event that denotes HEAD on the second throw.
Which of the following statements is/are TRUE?

- (A) A and B are independent.
- (B) A and C are independent.
- (C) B and C are independent.
- (D) $\text{Prob}(B|C) = \text{Prob}(B)$ CS

Question 11 of 23 | GATE 2022 Single Session | Q.28 (Computer Science & IT)

Q.28

Suppose a binary search tree with 1000 distinct elements is also a complete binary tree. The tree is stored using the array representation of binary heap trees. Assuming that the array indices start with 0, the 3rd largest element of the tree is stored at index_____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 12 of 23 | GATE 2022 Single Session | Q.57 (Computer Science & IT)

Q.57

Consider a network with three routers P, Q, R shown in the figure below. All the links have cost of unity.

The routers exchange distance vector routing information and have converged on the routing tables, after which the link Q–R fails. Assume that P and Q send out routing updates at random times, each at the same average rate. The probability of a routing loop formation (rounded off to one decimal place) between P and Q, leading to count-to-infinity problem, is_____.

P
Q
R

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 13 of 23 | GATE 2021 Set 1 (CS1) | Q.8 (Computer Science & IT)

Q.8

There are five bags each containing identical sets of ten distinct chocolates. One chocolate is picked from each bag.

The probability that at least two chocolates are identical is _____

- (A) 0.3024
- (B) 0.4235
- (C) 0.6976

(D) 0.8125

Question 14 of 23 | GATE 2019 Single Session | Q.20 (Computer Science & IT)

Q.20 An array of 25 distinct elements is to be sorted using quicksort. Assume that the pivot element is chosen uniformly at random. The probability that the pivot element gets placed in the worst possible location in the first round of partitioning (rounded off to 2 decimal places) is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 15 of 23 | GATE 2019 Single Session | Q.22 (Computer Science & IT)

Q.22 Two numbers are chosen independently and uniformly at random from the set $\{1, 2, \dots, 13\}$. The probability (rounded off to 3 decimal places) that their 4-bit (unsigned) binary representations have the same most significant bit is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 16 of 23 | GATE 2019 Single Session | Q.47 (Computer Science & IT)

Q.47 Suppose Y is distributed uniformly in the open interval $(1, 6)$. The probability that the polynomial $3x^2 + 6xY + 3Y + 6$ has only real roots is (rounded off to 1 decimal place) _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 17 of 23 | GATE 2018 Single Session | Q.15 (Computer Science & IT)

Q.15 Two people, P and Q , decide to independently roll two identical dice, each with 6 faces, numbered 1 to 6. The person with the lower number wins. In case of a tie, they roll the dice repeatedly until there is no tie. Define a trial as a throw of the dice by P and Q . Assume that all 6 numbers on each dice are equi-probable and that all trials are independent. The probability (rounded to 3 decimal places) that one of them wins on the third trial is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 18 of 23 | GATE 2018 Single Session | Q.44 (Computer Science & IT)

Q.44 Consider Guwahati (G) and Delhi (D) whose temperatures can be classified as high (H), medium (M) and low (L). Let $P(HG)$ denote the probability that Guwahati has high temperature. Similarly, $P(MG)$ and $P(LG)$ denotes the probability of Guwahati having medium and low temperatures respectively. Similarly, we use $P(HD)$, $P(MD)$ and $P(LD)$ for Delhi.

The following table gives the conditional probabilities for Delhi's temperature given Guwahati's temperature.

HD

MD
LD
HG
0.40
0.48
0.12
MG
0.10
0.65
0.25
LG
0.01
0.50
0.49

Consider the first row in the table above. The first entry denotes that if Guwahati has high temperature (HG) then the probability of Delhi also having a high temperature (HD) is 0.40; i.e., $P(HD|HG) = 0.40$. Similarly, the next two entries are $P(MD|HG) = 0.48$ and $P(LD|HG) = 0.12$. Similarly for the other rows.

If it is known that $P(HG) = 0.2$, $P(MG) = 0.5$, and $P(LG) = 0.3$, then the probability (correct to two decimal places) that Guwahati has high temperature given that Delhi has high temperature is _____.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 19 of 23 | GATE 2016 Set 1 (CS1) | Q.4 (Computer Science & IT)

Q.4

A probability density function on the interval $[a, 1]$ is given by $1/x^2$ and outside this interval the value of the function is zero. The value of a is

.

[Numerical Answer Type (NAT) Question — Enter Numerical Answer]

Question 20 of 23 | GATE 2016 Set 1 (CS1) | Q.29 (Computer Science & IT)

Q.29

Consider the following experiment.

Step 1. Flip a fair coin twice.

Step 2. If the outcomes are (TAILS, HEADS) then output Y and stop.

Step 3. If the outcomes are either (HEADS, HEADS) or (HEADS, TAILS), then output N and stop.

Step 4. If the outcomes are (TAILS, TAILS), then go to Step 1.

The probability that the output of the experiment is Y is (up to two decimal places)

.

CS(Set A)

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GATE 2016

*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 21 of 23 | GATE 2016 Set 2 (CS2) | Q.5 (Computer Science & IT)**

Q.5

Suppose that a shop has an equal number of LED bulbs of two different types. The probability of an LED bulb lasting more than 100 hours given that it is of Type 1 is 0.7, and given that it is of Type 2 is 0.4. The probability that an LED bulb chosen uniformly at random lasts more than 100 hours is

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*[Numerical Answer Type (NAT) Question — Enter Numerical Answer]***Question 22 of 23 | GATE 2016 Set 2 (CS2) | Q.22 (Computer Science & IT)**

Q.22

Suppose a database schedule S involves transactions $T_1, \dots, T_n$. Construct the precedence graph of S with vertices representing the transactions and edges representing the conflicts. If S is serializable, which one of the following orderings of the vertices of the precedence graph is guaranteed to yield a serial schedule?

- (A) Topological order
- (B) Depth-first order
- (C) Breadth-first order
- (D) Ascending order of transaction indices

Question 23 of 23 | GATE 2016 Set 2 (CS2) | Q.24 (Computer Science & IT)

Q.24

In an Ethernet local area network, which one of the following statements is TRUE?

- (A) A station stops to sense the channel once it starts transmitting a frame.
- (B) The purpose of the jamming signal is to pad the frames that are smaller than the minimum frame size.
- (C) A station continues to transmit the packet even after the collision is detected.
- (D) The exponential backoff mechanism reduces the probability of collision on retransmissions. CS(Set B) 6/18
GATE 2016